P.T. INTI KARYA PERSADA TEHNIK ENGINEERING & CONSTRUCTION KELLOGG OVERSEAS CORPORATION	DOC. No. DS-H04-K-002 Job No. : 95036 / 6413	SHEET No. : 1 OF 16

DATA SHEET :

MCR FIRST STAGE COMPRESSOR
H4K-2

4		1/5/97
REV. NO	APPROVED BY	DATE
FINAL APPROVAL BY PKP		

CLIENT : PERUSAHAAN PERTAMBANGAN MINYAK DAN GAS BUMI NEGARA (PERTAMINA)/ PROYEK KILANG PENGOLAHAN (PKP)

JOB TITLE : BONTANG LNG EXPANSION PROJECT TRAIN H

PROJECT NAME : PGH PROJECT

LOCATION : BONTANG, EAST KALIMANTAN, INDONESIA

FOR RECORD

REV	DATE	PAGE	DESCRIPTION	PREP'D	CHK'D	APP'D
0	19 Dec.1996	ALL	Issued for Approval and Inquiry	ADN SDP	AS	BS JGB
1	29 Jan. 1997	ALL	For PKP review during Bid Clarification Meeting	ADN SDP	AS SWK	
2	6 Feb. 1997	ALL	Issued for Approval based on PKP comments at Bid Clarification Meeting	ADN IRA	AS SWK	BS JGB
3	24 Mar. 1997	ALL	Issued for PKP Approval revised based on PKP comments HX-PJIP-0049	ADN	AS SWK	BS JGB
4	16 Apr. 1997	—	Approved per HX-PJIP-0112/0165 and issued for Purchase Order	ADN	AS SWK	JGB

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CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-2
P.O. NO. DATE 24 March 1997
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-002

APPLICABLE TO : ○ PROPOSAL ◉ PURCHASE ○ AS BUILT
DATE 24 February 1997 REVISION 3
FOR BONTANG LNG EXPANSION PROJECT, TRAIN-H
UNIT PLANT 4 - REFRIGERATION
SITE BONTANG, EAST KALIMANTAN, INDONESIA
SERIAL NO.
SERVICE MR FIRST STAGE COMPRESSOR
NO. REQUIRED ONE
MANUFACTURER DRESSER - RAND MODEL 838 B4
DRIVER STEAM TURBINE (H4KT-2)
NOTE : INFORMATION TO BE COMPLETED : ○ BY PURCHASER ☐ BY MANUFACTURER.

OPERATING CONDITIONS

(ALL DATA ON PER UNIT BASIS)

	NORMAL & RATED			ALT. OPERATION		
	CASE 1			CASE 2	CASE 3	Recycle (2) by Vendor
◉ GAS HANDLED (Also See Page 2) nm3/h (1.013 bar & 0°C Dry)	712268			705844	703421	
◉ Weight Flow, kg/h (Dry)	759,018 x 1.025			751,221	748,735	
INLET CONDITIONS						
◉ Pressure, (Kg/cm2) A NOTE (3)	3.43			3.40	3.38	
◉ Temperature, °C NOTE (3)	-38.3			-37.13	-37.02	
○ Mix Temperature °C						
◉ Molecular Weight, M	25.075			25.044	25.046	
◉ Cp / Cv (K1)	1.229			1.228	1.228	
☒ Compressibility (Z1)	.967			.968	.969	
☒ Inlet Volume, m3/h (Dry)	174277			175315	175835	
DISCHARGE CONDITIONS						
☒ Pressure, (Kg/cm2) A NOTE (4)	15.84			15.69	15.62	
☒ Temperature, °C (to be verified by vendor)	61.4			62.8	63	
☒ Cp / Cv (K2) AVG	1.229			1.228	1.228	
☒ Compressibility (Z2) AVG	.961			.962	.962	
☒ KW Required (All Losses Included) ± 4 %	36754		→	36667	36614	
☒ Speed, rpm	4610		→	4631	4639	
☐ Estimated Surge, m3/h (at Speed Above)						
☒ Polytropic Head, M-kg/kg	14000			14100	14119	
☒ Polytropic Efficiency, %	80.5			80.5	80.4	
◉ Guarantee Point See note 1 , page 14 of 16	YES					
☒ Performance Curve No.	see pages	3, 4, 5, 6, 7, 8 of 16				→

PROCESS CONTROL By HCJV

Method : ○ Suction Throttling From ____ (bar) To ____ (bar) ○ Variable Inlet Guide Vanes (3.4.2.4) ◉ Speed Variation From 80 % To 105 % ○ Discharge Blowoff To ____ ◉ Cooled Bypass From Discharge To Suction

Signal : ◉ Source (3.4.2.1) MANUAL CONTROLLER (Electronic Signal/hydraulically Actuated) FOR SPEED CONTROL

Type : ○ Electronic ○ Pneumatic ◉ Other ANTI-SURGE CONTROLLER (Electronic Signal / Pneumatically Actuated) FOR COOLED BY PASS

Range 4-20 mA 0.2-1.0 (kg/cm2G)

Antisurge Bypass : ○ Manual ◉ Automatic ○ None

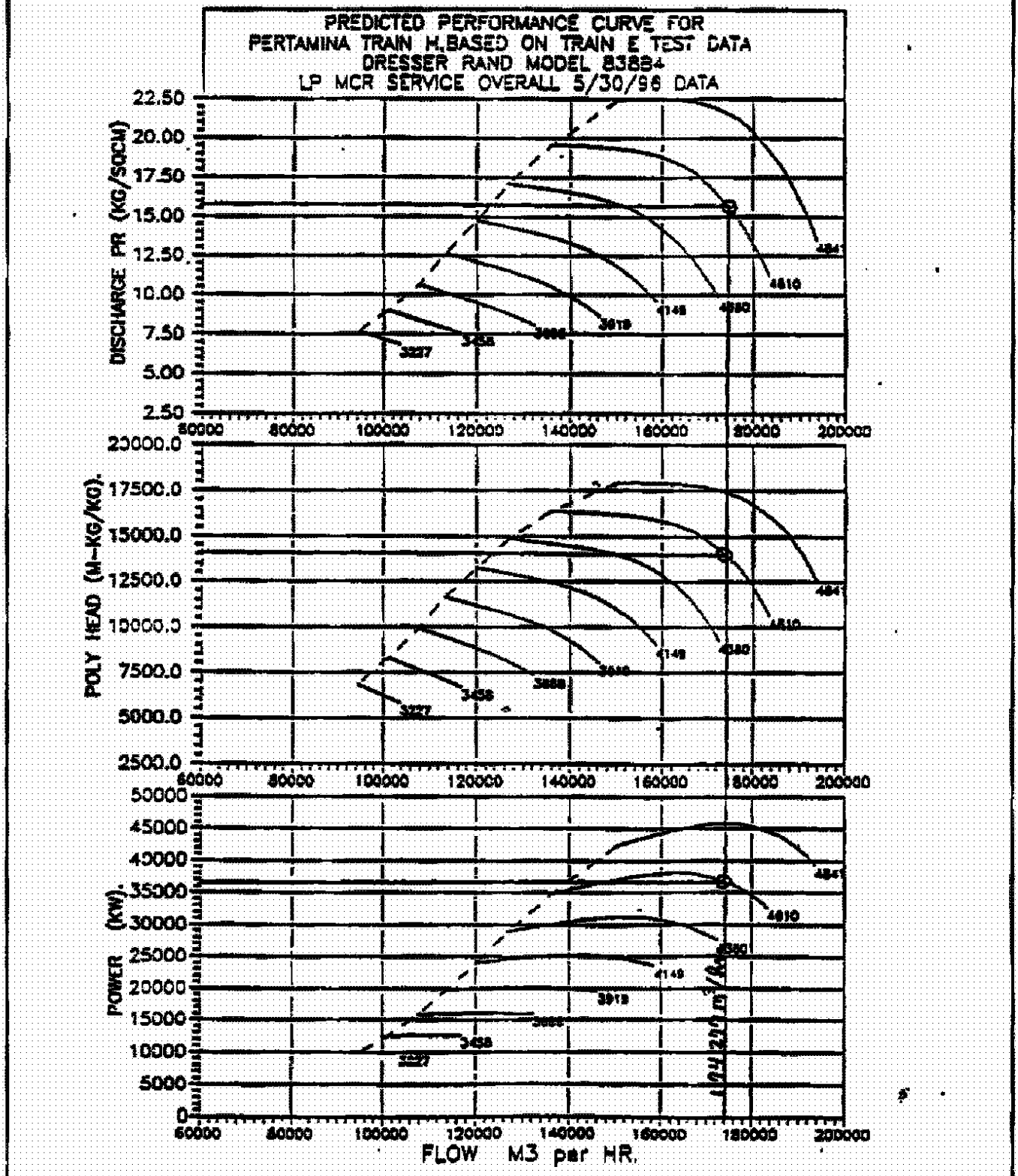
REMARKS

(1) VENDOR SHALL CONFIRM INLET AND DISCHARGE GAS PROPERTIES.
(2) VENDOR SHALL CONFIRM COMPRESSOR PERFORMANCE DURING FULL RECYCLE. GAS TEMPERATURE UPSTREAM OF ANTI-SURGE CONTROL VALVE IS 32.7°C AT 15.44 KG/CM2 (ABS), 15.29 KG/CM2 (ABS), AND 15.22 KG/CM2 (ABS) FOR CASES 1, 2 AND 3 RESPECTIVELY. RECYCLE GAS TO BE EXPANDED TO SUCTION PRESSURE OF EACH RESPECTIVE CASE.
(3) VENDOR TO COMPLETE THE DATA IN THE BLANK CELLS

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CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-2
P.O. NO. _____ DATE 24 March 1997
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-002



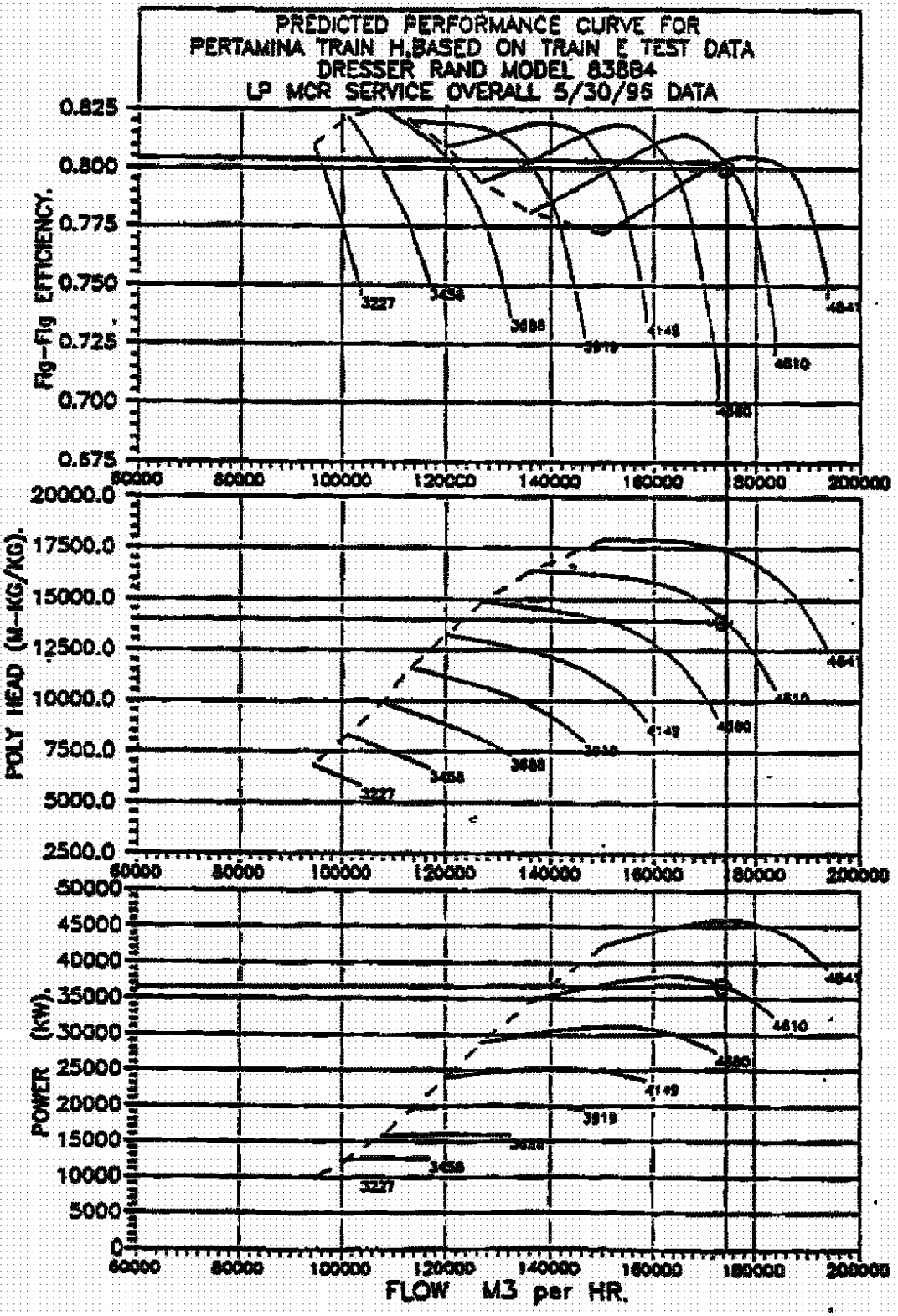
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NS-IN0224		LPMCRO

DRESSER-RAND

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CENTRIFUGAL COMPRESSOR
DATA SHEET
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JOB NO. 95036/6413 ITEM NO. H4K-2
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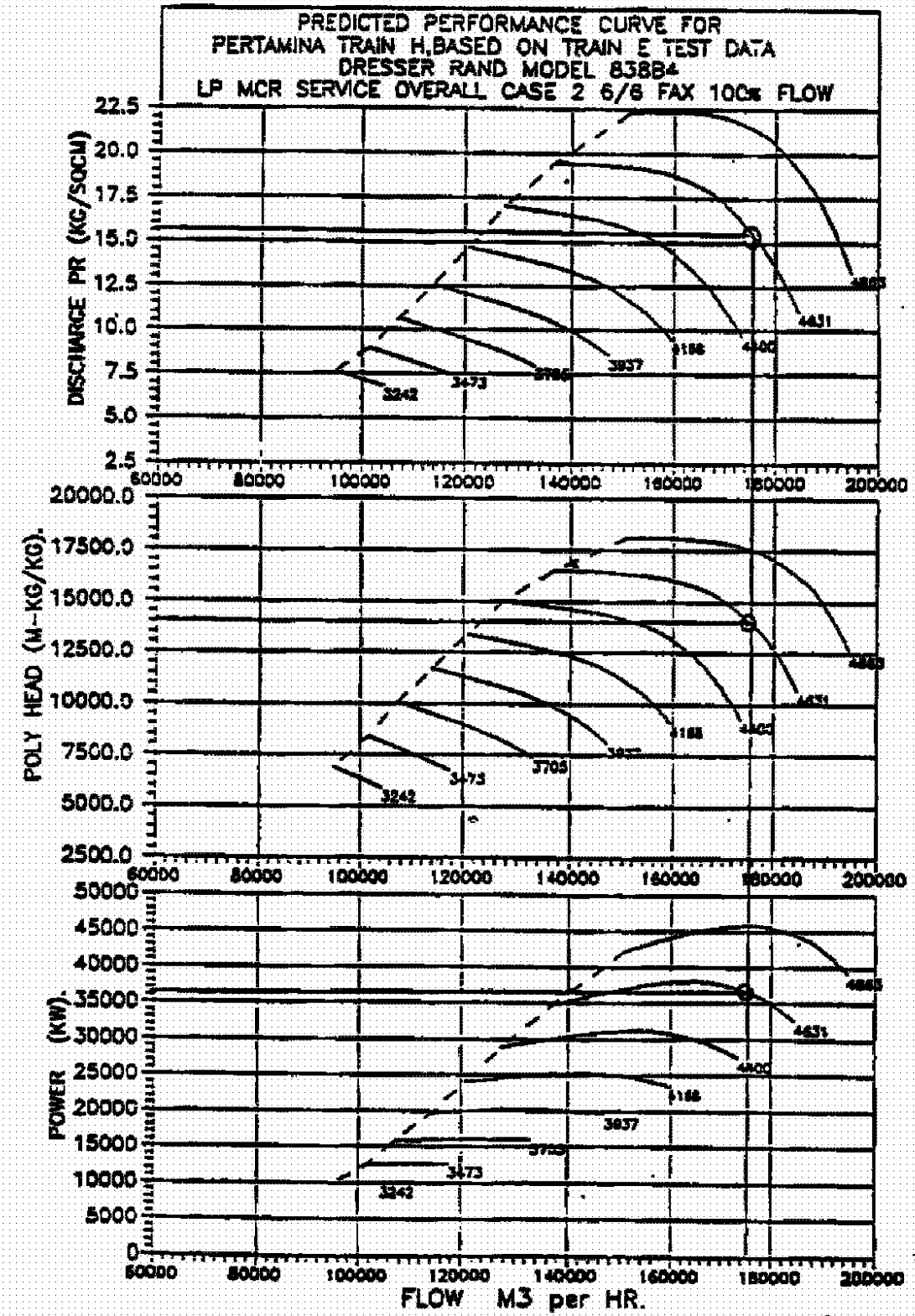
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NS-IN0224		LPMCRDA

DRESSER-RAND

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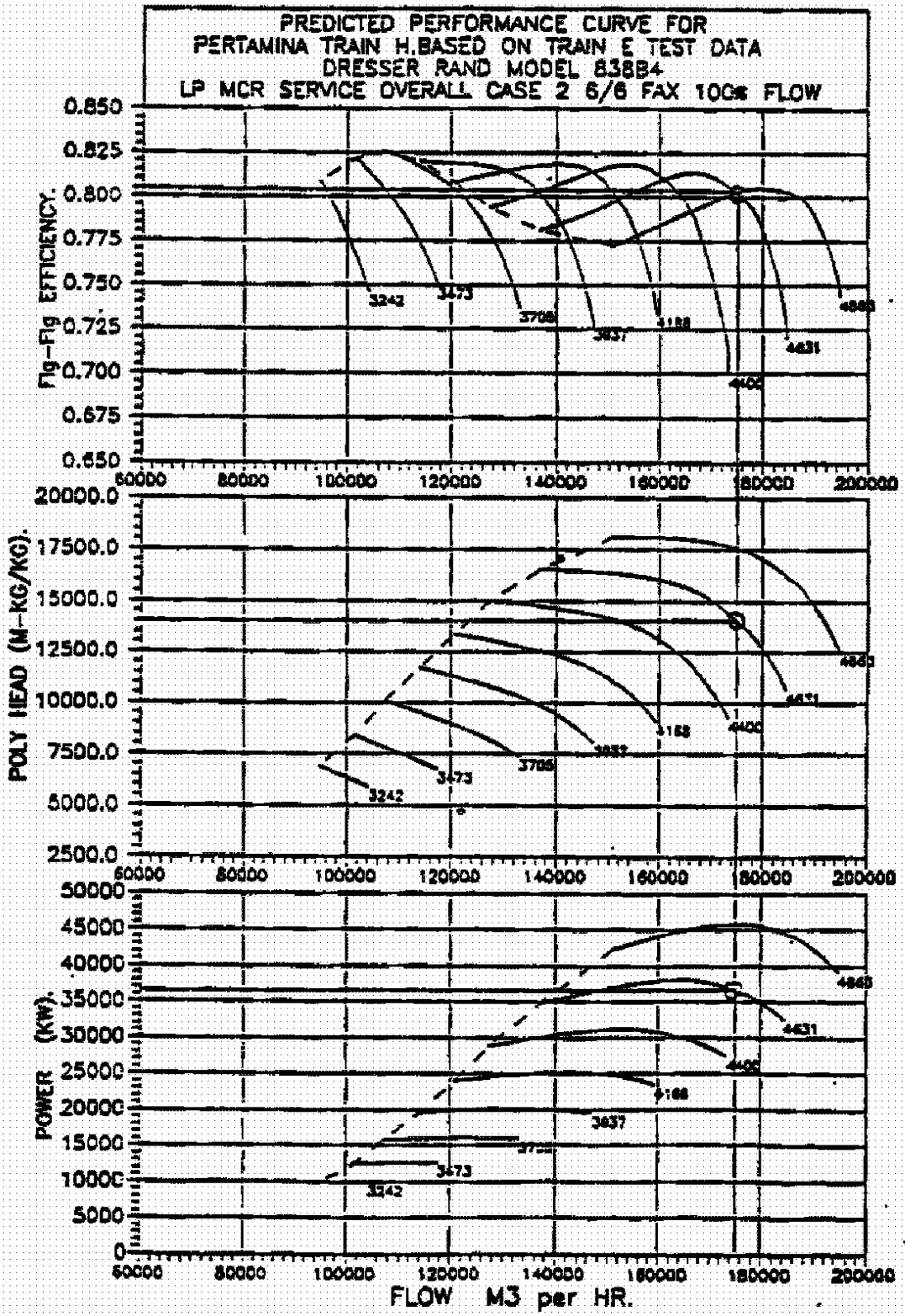
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N5-tN0024		LPMCR20

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CENTRIFUGAL COMPRESSOR
DATA SHEET
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JOB NO. 95036/6413 ITEM NO. H4K-2
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DOC. NO. DS-H04-K-002



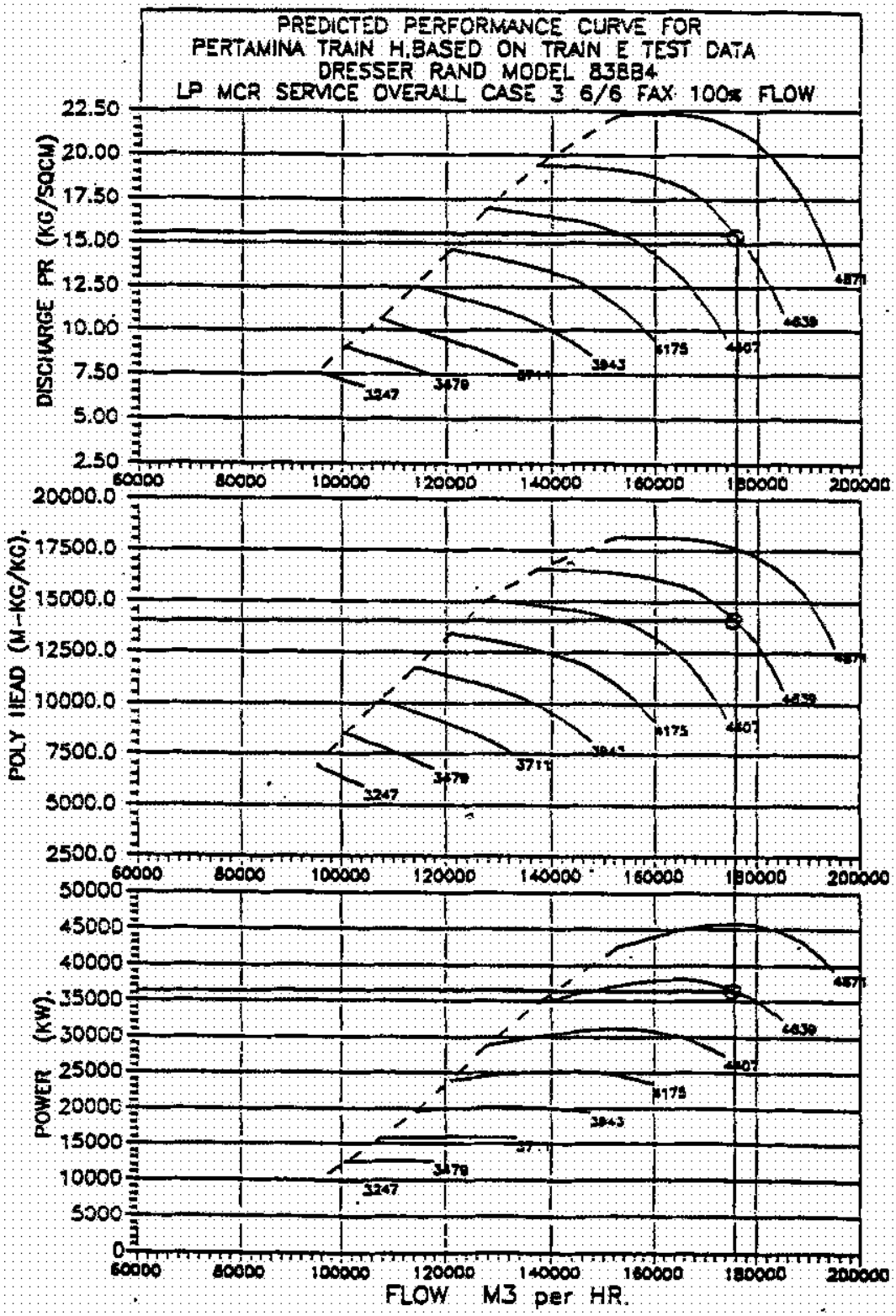
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CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. *95038/6413* ITEM NO. *H4K-2*
P.O. NO. _____ DATE *24 March 1997*
REQUISITION NO. *PGH-MR-K-003*
DOC. NO. *DS-H04-K-002*



DATE: 6-6-96	TTB	CURVE No.:
N5-IN:0224		LPMCR30

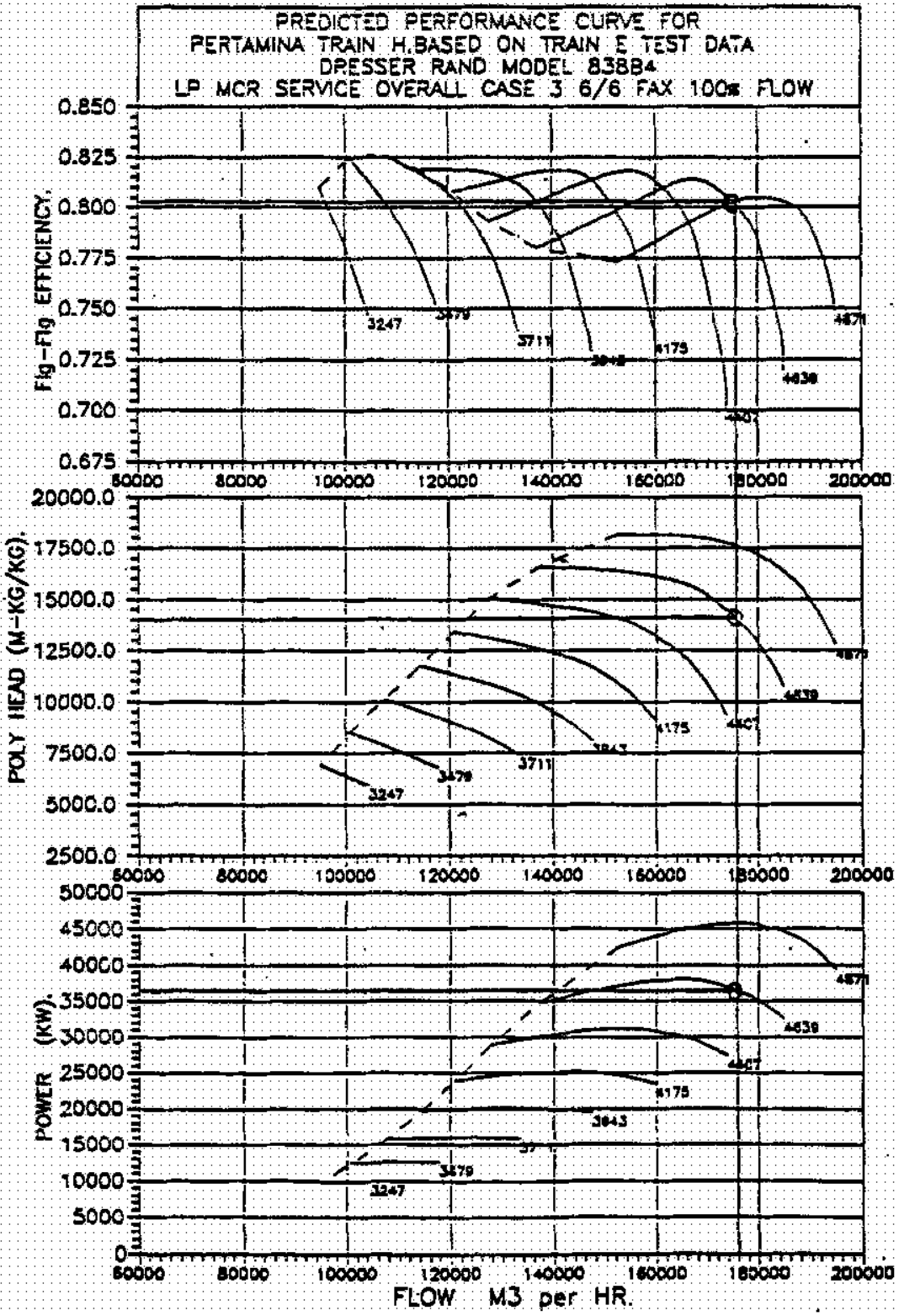
DRESSER-RAND

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CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. *95038/6413* ITEM NO. *H4K-2*
P.O. NO. DATE *24 March 1997*
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DOC. NO. *DS-H04-K-002*



DATE: 6- 6-96	TTB	CURVE No.:	
N5-IN0224		LPMCR30A	

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CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

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DOC. NO. DS-H04-K-002

OPERATING CONDITIONS (CONTD) (3.1.2 & 3.1.3)

GAS ANALYSIS ◉ MOL % ○	MW	DESIGN & GUARANTEE CASE 1	CASE 2	CASE 3	REMARKS (2.11.1.3 & 2.11.1.8)
Air	28.966				
Oxygen	32.000				
Nitrogen	28.016	3.60	3.45	3.32	
Water Vapor	18.016				
Carbon Monoxide	28.010				
Carbon Dioxide	44.010				
Hydrogen Sulfide	34.076				
Hydrogen	2.016				
Methane	16.042	45.16	44.75	46.01	
Ethylene	28.052				
Ethane	30.068	41.16	42.37	39.98	
Propylene	42.078				
Propane	44.094	10.08	9.43	10.69	
Iso Butane	58.120				
n Butane	58.120				
Iso Pentane	72.146				
n Pentane	72.146				
Hexane Plus					
TOTAL		100	100	100	
Avg Mol Wt		25.075	25.044	25.046	

LOCATION
○ Indoor ○ Heated ◉ Under Roof
◉ Outdoor ○ Unheated ◉ Partial Sides
○ Grade ◉ Mezzanine ○ ____
◉ Electrical Area Class. (2.1.15) Cl 1 Gr D Div 2
○ Winterization Reqd (2.1.9) ◉ Tropicalization Reqd (3.4.6.6)

NOISE SPECIFICATIONS (2.1.10)
◉ Applicable to Machine
See Specification SPEC. A-9
○ Applicable to Neighborhood
See Specification
Acoustic Housing : ○ YES ◉ NO

SITE DATA
◉ Elevation 14 m ◉ Barometer 1.03 Kg/cm2
◉ Range of Ambient Temp,° C :

	DRY BULB	WET BULB
Site Rated	30°C	28°C
Normal	26°C	24°C
Max.	37°C	-
Min.	21°C	-

APPLICABLE SPECIFICATIONS API Std 617
◉ Vendor Having Unit Responsibility (2.9.1.7)
◉ Governing Specification
SPEC. K-1 (C.2.7.1)

PAINTING
○ Manufacturer's ◉ Other SPEC. X-2 & PGG-IS-X-1002 REV. 2

UNUSUAL CONDITIONS ○ Dust ○ Fumes
◉ Other (2.1.9)
SALT LADEN ATMOSPHERE

SHIPMENT (4.4.1)
○ Domestic ◉ Export ◉ Export Boxing Reqd
◉ Outdoor Storage More Than 6 Months (4.4.1)
Spare Rotor Assembly Packaged for (4.4.3.10)
○ Horizontal Storage ○ Vertical Storage

REMARKS
(4) CASE 1 : MINIMUM BUTANE RECYCLE
CASE 2 : MAXIMUM BUTANE RECYCLE
CASE 3 : NO LPG PRODUCTION

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CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-2
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CONSTRUCTION FEATURES

☒ SPEEDS
Max. Cont 4873 rpm Trip 5360 rpm
Max. Tip Speeds : 308.7 m/s @ rated Speed
324 m/s Max. Cont Speed

☒ LATERAL CRITICAL SPEEDS (DAMPED)
First Critical 2500 rpm Mode
Second Critical 8800 rpm Mode
Third Critical rpm Mode
Fourth Critical rpm Mode
• Lateral Analysis Required
• Undamped Stiffness Map Required
• Train Torsional Analysis Required
(Turbine Driven Train) (2.9.4.5)

☐ TORSIONAL CRITICAL SPEEDS
First Critical rpm
Second Critical rpm
Third Critical rpm
Fourth Critical rpm

☒ VIBRATION
Allowable Test Level (pk-to-pk) 38.1 Microns

ROTATION, VIEWED FROM DRIVEN END

• MATERIALS INSPECTION REQUIREMENTS (4.2.2)
• Special Charpy Testing (2.11.3)
• Radiography Required for (See Note 2 Pg. 14)
• Magnetic Partical Required for (See Note 2 Pg. 14)
Liquid Penetrant Required for

☒ CASING
Model 838 B4
Casing Split RADIAL
Material ASTM A 203 Gr E
Thickness, mm 69.85 Corr. Allow., mm 3.18
Max. Working Press. 20.5 kg/cm2G
Max. Design Press. 28.1 kg/cm2G
Test Press., Kg/cm2G: Helium N/A Hydro 42.2
Max. Oper Temp (1) 204.4 °C Min. Oper Temp. - 59.4 °C
Max. No. of Impellers for casing FOUR
Max. Casing Capacity, m3/h 220000
Radiograph Quality • Yes ○ No
Casing Split Sealing O - RINGS
• System Relief Valve Set Pt (2.2.4) 20.5 kg/cm2G

☒ DIAPHRAGM
Material FABRICATED STEEL

☒ IMPELLERS
No. 4 Diameters 1270, 1207, 1219, 1080
No. Vanes Each Impeller 17, 21, 17, 19
(1) VENDOR SHALL CONFIRM BASED ON CONTINUOUS RECYCLE CASE

Type (Open, Enclosed, etc) ENCLOSED
Type Fabrication WELDED
Material AISI 4330
Min. Yield Strength, Kg/cm2 10900 MIN
Brinnel Hardness : Max. 375 Min. 332
Smallest Tip Internal Width, mm 78
Max. Mach No. @ Impeller Eye 76
Max. Impeller Head @ Rated Speed, m 5112

☒ SHAFT
Material AISI 4340
Diam @ Impellers, mm 1286 Diam @ Coupling, mm 208
Shaft End : ☒ Tapered Degree of Angle
Max. Yield Strength, bar 5862
Shaft Hardness (BNH) (Rc) 217 - 277
Stress at Coupling, bar 479

☒ BALANCE PISTON
Material 4140 Area 3677 mm2
Fixation Method INTERFERENCE FIT

☒ SHAFT SLEEVES
At Interstg Close Clearance Points Matl 410 SS
At Shaft Seals Matl MONEL "B" WITH COLMONOY OVERLAY

☒ LABYRINTHS
Interstage :
Type KNIFE EDGE Material ALUMINIUM
Balance Piston :
Type KNIFE EDGE Material ALUMINIUM

SHAFT SEALS
◉ Seal Type (2.8.3) OIL FILM
◉ Settling Out Pressure (150 PSIG) 10.55 kg/cm2G
○ Special Operation (2.8.1)
○ Supplemental Device Required for Contact Seals (2.8.3.2)
Type
○ Buffer Gas System Required (2.8.7)
○ Type Buffer Gas
○ Buffer Gas Control System Schematic by Vendor
☐ Type Seals
☒ Inner Oil Leakage Guar. L/day/seal 428 TOTAL
Buffer Gas Required :
☐ Air-Run-in ☐ Other
☐ Buffer Gas Flow (per Seal) :
Norm : kg/min @ bar ΔP
Max. : kg/min @ bar ΔP

☒ BEARING HOUSING CONSTRUCTION
Type (Separate, Integral) SEPARATE Split HORIZONTAL
Material C.S ASTM A 36 Gr D
Separation Seal, Type Split
Material

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CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-2
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DOC. NO. SH-DS-04-K-02

CONSTRUCTION FEATURES (CONTD)

BEARINGS AND BEARING HOUSINGS

RADIAL

	INLET	EXHAUST
Type	Tilting Pad	Tilting Pad
Manufacturer	D-R	D-R
Length, mm	92	92
Shaft Diam, mm	228.6	228.6
Unit Load (Act. / Allow.), bar	6.57 / 34.5	6.57 / 34.5
Base Material	ASTM A36	ASTM A36
Babbit Thickness, mm	1.524	1.524
No. Pads	FIVE	FIVE
Load : Between / On Pad	ON	ON
Pivot : Center / Offset, %	CENTER	CENTER
Bearing Span	2737	

THRUST

	ACTIVE	INACTIVE
Type	Tilting Pad	Tilting Pad
Manufacturer	D-R	D-R
Unit Loading, Max. bar	7.3	7.3
Unit Load, Ult bar	69	69
Area, mm2	72516	72516
No. Pads	12	12
Pivot : Center / Offset, %	55	55
Pad Base Material	ASTM A36	ASTM A36

Lubrication : (•) Flooded ( ) Directed
Thrust Collar : ( ) Integral (•) Replaceable
Material FORGED STEEL

BEARING TEMPERATURE DEVICES

( ) See Attached API 670 Data Sheet
( ) Thermistors
( ) Type - Pos Temp Coeff - Neg Temp Coeff
( ) Temp Switch & Indicator by : - Purch - Mfr
(•) Thermocouples
(•) Selector Switch & Indicator by : √ Purch - Mfr
( ) Resistance Temp Detectors
( ) Resistance Matl - - Ohms
( ) Selector Switch & Indicator by : - Purch - Mfr
(•) Location-Journal Bearing
No. - Each Pad - Every Other Pad 4* per Brg
Other (*) Include 2 Spares
(•) Location-Thrust Bearing
No. - Each Pad - Every Other Pad 4** per Brg
Other (**) Include 2 Spares
No. (Inact) - Each Pad - Every Other Pad 2 per Brg
Other
(•) Monitor Supplied by (3.4.7.4) PURCHASER
( ) Location - Enclosure -
( ) Mfr - [ ] Model -
[ ] Scale Range - (•) Alarm [X] Set @ 110 °C
3 (•) Shutdown [ ] Set @ 121 °C ( ) Time Delay - sec

VIBRATION DETECTORS

( ) See Attached API 670 Data Sheet
(•) Type PROXIMITY [X] Model 3300
(•) Mfr BENTLY NEVADA
(•) No. at Each Shaft Bearing 4*** Total No. 8
(•) Oscillator-Detectors Supplied by COMPRESSOR MFR.
(•) Mfr BENTLY NEVADA [X] Model 3300
Monitor Supplied by (3.4.7.2) PUCHASER
(•) Location CCR Enclosure
(•) Mfr BENTLY NEVADA [X] Model 3300
[ ] Scale Range (•) Alarm [X] Set @ 102 mm
(•) Shutdown [X] Set @ 132 mm ( ) Time Delay sec
3

AXIAL POSITION DETECTOR

( ) See Attached API 670 Data Sheet
(•) Type PROXIMITY [X] Model 3300
(•) Mfr BENTLY NEVADA (•) No. Required 3****
(•) Oscillator-Demodulator Supplied by COMPR. MFR.
(•) Mfr BENTLY NEVADA [X] Model 3300
Monitor Supplied by (3.4.7.2) PURCHASER
(•) Location CCR Enclosure
(•) Mfr BENTLY NEVADA [X] Model 3300
[ ] Scale Range (•) Alarm [X] Set @ 127 mm
(•) Shutdown [X] Set @ 254 mm ( ) Time Delay sec
3

CASING CONNECTIONS

1 CONNECTION	(•) DESIGN APPROVAL REQD (2.11.2.10) 0	[X] SIZE	[X] FACING	(•) POSITION (2.4.2.1)	(•) FLANGED OR STUDDED (2.4.2.1)	(•) MATING FLG & GASKED BY VENDOR (2.4.2.3)	[X] GAS VELOCITY @ MCS m/s Rtd / Max. Allow.
Inlet	Yes	48	Taylor 350#RF	DOWN	FLANGED	Yes	41.85
Discharge	Yes	30	MSS 300# RF	DOWN	FLANGED	Yes	33.6

(***) INCLUDES 2 SPARES INSTALLED
(****) INCLUDES 1 SPARE INSTALLED

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☐ OTHER CONNECTONS

SERVICE	NO	SIZE	TYPE
Lube Oil Inlet			
Lube Oil Outlet			
Seal Oil Inlet			
Seal Oil Outlet			
Casing Drains			
Stage Drains			
Vents			
Cooling Water			
Pressure			
Temperature			
Purge for :			
Brg Housing			
Between Brg & Seal			
Buffer Gas			
Solvent Injection			

◉ Individual Stage Drains Required (2.4.3.2)
○ Valved & Blinded
◉ Valved & Blinded & Manifold

☒ ALLOWABLE PIPING FORCES AND MOMENTS

	INLET FORCE N	INLET MOMT N.m	DISCHARGE FORCE N	DISCHARGE MOMT N.m	FORCE N	MOMT N.m
AXIAL						
VERTICAL	PER	API	617	6TH	ED.	
HORIZ 90°						

	FORCE N	MOMT N.m	FORCE N	MOMT N.m	FORCE N	MOMT N.m
AXIAL						
VERTICAL	PER	API	617	6TH	ED.	
HORIZ 90°						

○ ACCELEROMETER
○ See Attached API 670 Data Sheet
○ Type ☐ Model
○ Mfr No. Required
○ Location
○ Oscillator-Demodulators Supplied by
○ Mfr ☐ Model
○ Monitor Supplied by (3.4.7.6)
○ Location Enclosure
○ Mfr ☐ Model
☐ Scale Range ○ Alarm : ☐ Set @ mm/s2
○ Shutdown ☐ Set @ mm/s2 ○ Time Dela sec

ACCESSORIES

COUPLING AND GUARD (3.2)

NOTE : See Rotating Elements - Shaft Ends
○ See API 671 Data Sheet
Coupling Furnished by Compressor Mfr.
Manufacturer Lucas (Bendix) Type Countoured Diaphragm Flexible Model 74 T522 A
Coupling Guard Furnished by Compressor Mfr No. of Vent Conn : 1 Size, mm 203.2
Type : ◉ Fully Enclosed ○ Semi-open ○ Oth

Coupling Details :
☐ Max. OD mm
☐ Hub Weight kg
☐ Spacer Length mm
☐ Spacer Weight kg

◉ Vendor Mount Half Coupling
○ Idling Adapter / Solo Plate Reqd (3.2.4)
Lubricating Requirements :
◉ Nonlube ○ Grease ○ Cont Oil Lube ○ Other
Quantity per Hub kg or m3/h
◉ Hydraulic coupling removal / attachment KIT
(Adapter + Hydraulic Pump)

MOUNTING PLATES

◉ Baseplates : Furnished by (3.3.1.1) Compressor Mfr
◉ Compressor Only (3.3.2.1) ○ Driver ○ Gear
○ Other
○ Drip Trim ○ Leveling Pads (3.3.2.2)
○ Column Mounting (3.3.2.3)
○ Subsoleplates Reqd (3.3.3.2)
☒ Stainless Steel Shim Thickness 6.3 mm
◉ Primer for Epoxy Grout Reqd (3.3.1.2.9)
Type Catalyzed Epoxy Primer

○ Soleplates : Furnished by
☐ Thickness mm
○ Subsoleplates Reqd (3.3.3.2)
○ Levelling (Chock) Blocks Reqd
☐ Stainless Steel Shim Thicknes
○ Driver ○ Gear ○ Compressor
○ Primer for Epoxy Grout Reqd (3.3.1.2.9)
Type

REMARKS

IKPT — KELLOGG

CENTRIFUGAL COMPRESSOR DATA SHEET SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-2
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UTILITIES

○ UTILITY CONDITIONS

STEAM	DRIVERS		HEATING	
Inlet Min.	17.6 kg/cm2	°C	barg	°C
Norm.	17.6 kg/cm2	°C	barg	°C
Max.	19.3 kg/cm2	°C	barg	°C
Exhaust Min.	3.9 kg/cm2	°C	barg	°C
Norm.	3.9 kg/cm2	°C	barg	°C
Max.	3.9 kg/cm2	°C	barg	°C

ELECTRICITY	DRIVERS	HEATING	CONTROL	SHUTDOWN
Voltage	460	240	120	120
Hertz	50	50	50	50
Phase	3	1	1	1

COOLING WATER

Temp : Inlet 34 °C — Max. Return 46 °C
Press. : Norm. 2.1 Kg/cm2G — Design 10.5 Kg/cm2G
Min. Return 1.1 Kg/cm2G — Max. Allow. ΔP 1.0 Kg/cm2G
Water Source See Water

INSTRUMENT AIR

Max. Press. 9.0 Kg/cm2G — Normal Press. 7.0 Kg/cm2G — Min. Press. 4.2 Kg/cm2G

SHOP INSPECTION AND TEST (4.1.4)

	REQD	WITNESS	OBSERV'D
Shaft Runout	●	●	
Shop Inspection	●	●	
Cleanlines (4.2.1.5)	○		
QC Program Review	●		
Hydrostatic	●	●	○
Impeller Overspeed	●	●	○
Mechanical Run *with shop driver*	●	●	○
☒ Contract Coupling ☐ Idling Adaptor(s)			
☒ Contract Probes ☐ Shop Probes			
Vary Lube & Seal Oil Pressures and Tempeatures (4.3.4.2.5)	●		
Polar Form Vibration Data (4.3.4.3.3)	●		
Tape Record Vibration Data (4.3.4.3.6)	●		
Tape Data to Purchaser (4.3.4.3.7)	●		
Shaft End Seal Inspection (4.3.4.4.1)	●	●	
Gas Leak Test Disch Press. (4.3.5.3)	●	●	○
○ Before ◉ After Mech Run			
Performance Test (Gas) (4.3.6.1)	●	●	○
Complete Unit Test (4.3.6.2)	○	○	○
Torsional Vibration Meas (4.3.6.2)	○	○	○
Tandem Test (4.3.6.3)	○	○	○
Gear Test (4.3.6.4)	○	○	○
Helium Leak Test (4.3.6.5)	○	○	○
Sound Level Test (4.3.6.6)	●	●	○
Full Load/Speed/Press. Test (4.3.6.9)	○	○	○
Hydraulic Coupling Inspection (4.3.6.10)	●	●	○
Certified Copies of Test Data	●	○	○
Disassy. / Reassy. Compressor After Test (3)	●	●	○
Preparation for Shipment	●	●	○
Residual Elect / Mech Run Out	●	●	○
Bearing / Seal Disassembly	●	●	
Unbalance Response Verification	●	●	

☐ TOTAL UTILITY CONSUMPTION

Cooling Water	m3/h
Steam, Normal	Kg/h
Steam, Max.	Kg/h
Instrument Air.	mm3/h
Horsepower (Driver)	k/W
Horsepower (Auxiliaries)	k/W
Heaters	k/W
Purge (Air or N2)	mm3/h

MISCELLANEOUS

☒ Recommended Straight Run of Pipe Diameters
◉ Before Suction THREE
◉ Vendor's Review & Comments on Purchaser's Piping & Foundation (3.5.3.2)
○ Compressor to be Suitable for Field Run-in on Air
○ Provision for Liquid Injection (2.1.11)

◉ Vendor's Review & Comments on Purchaser' Control System (3.4.1.1)
○ Extent of Process Piping by Vendor (3.5.3.1)
○ Shop Fitup of Vendor Process Piping (4.4.3.11)
○ Welding Hardness Testing (4.2.1.6)
○ Auxiliary Equipment Motors Explosionproof (3.1.8)
◉ Torsional & Lateral Critical Analysis Report Req'd.

☒ WEIGHTS, kg

Compr 45600 Gear — Driver 47200 Base 7800
Rotors : Compr 2800 Driver 8900 Base —
Compr Upper Case
Sour Seal Oil Traps
L. O. Console 37700 S. O. Console
Overhead Seal Oil Tanks
Max. for Maintenance (Identify) 21000
Total Shipping Weight 53400 Comp. 47200 S.Turbine

☒ SPACE REQUIREMENTS, m

Complete Unit : L 13600 W 4030 H 4700
L. O. Console : L 11500 W 5600 H 2700
S. O. Console : L W H
Sour Seal Oil Tanks
Overhead Seal Oil Tanks

SAFETY CONTROLS

Control Logic : C
— Relay — Single — Dual — 2 of 3
— PLC — Dual — 2 Out of 3
Vendor to Privide :
— Total System — Local Components
PLC :
Interface — Mfr

REMARKS

1. UTILITY CONDITIONS, REFER TO SH-ED-PE-03 (1)
2. OTHER SHOP INSPECTION AND TESTS SHALL BE BE CONDUCTED AS PER PGH-QC-A-002.
3. IF UNSATISFACTORY RUN TEST (2)

IKPT KELLOGG

CENTRIFUGAL COMPRESSOR
DATA SHEET
SI (METRIC) UNITS

JOB NO. 95038/6413 ITEM NO. H4K-2
P.O. NO. _____ DATE 24 March 1997
REQUISITION NO. PGH-MR-K-003
DOC. NO. DS-H04-K-002

SPECIAL NOTES ;

(1) THE ACCEPTANCE CRITERIA FOR THE SHAFT HORSE POWER INCLUDING ALL LOSSES AT THE GUARANTEE POINT (CASE 1) SHALL NOT EXCEED 104% OF THE SPECIFIED VALUE DURING THE PTC-10 SHOP PERFORMANCE TEST.

(2) MATERIAL TEST, SHOP INSPECTION AND TEST REQUIREMENTS OTHER THAN THOSE SPECIFIED IN THIS DATA SHEET SHALL BE REFERRED TO PGH-QC-A-002 " SHOP INSPECTION AND TEST PLAN".

(3) COMPRESSOR AND TURBINE VENDOR SHALL DEMONSTRATE INTERCHANGEABILITY OF THE COMPLETE BUNDLE (ROTOR, DIAPHRAGMS, ETC) WITH EXISTING 4K-2 CASES (IE. IN THE EXISTING TRAIN G) . VENDOR TO PROVIDE A DIMENSIONAL RECORD TO HCJV (PURCHASER) FOR REVIEW AND APPROVAL.

(4) SEAL AND BEARING PARTS MUST BE INTERCHANGEABLE WITH THE EXISTING 4K-2's TRAIN-G

(5) SHAFT COUPLING SHALL BE RATED ACCORDING TO THE POWER REQUIREMENTS SPECIFIED ON PAGE 2.

(6) OTHERS:

a. TORSIONAL AND LATERAL CRITICAL ANALYSIS REPORT REQUIRED WITH COMPLETE ROTOR DYNAMICS AND BEARING STABILITY REPORT PER SPECIFICATION

b. VENDOR TO TERMINATE ALL WIRING AT TERMINAL BOXES AT EDGE OF SKID

c. OPTICAL ALIGMENT PADS (BENCH MARKS) REQUIRED ON COMPRESSOR / DRIVER SKID

2 *d. DELETED*

IKPT

KELLOGG

COUPLING (API 671-2ND)
DATA SHEET
SI UNITS

JOB NO. 95038 / 6413 ITEM NO. H4K-2
PURCHASE ORDER NO. PGH-PO-K-003
REQUISITION NO. PGH-MR-K-003
REVISION NO. 3 DATE 24 MARCH 1997
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APPLICABLE TO: ○ PROPOSAL ● PURCHASE ○ AS BUILT DATE 6 FEB '97 REVISION 3

● FOR BONTANG LNG EXPANSION PROJECT, TRAIN - H NO REQUIRED 1 (ONE)
SITE BONTANG, EAST KALIMANTAN, INDONESIA SERVICE LP MR COMPRESSOR

● DRIVER TYPE STEAM TURBINE MANUFACTURER GENERAL ELECTRIC MODEL DRV 1034
SERIAL NO. TAG NO.
DRIVEN UNIT COMPRESSOR MANUFACTURER DRESSER RAND MODEL 838 B4
SERIAL NO. TAG NO.

■ COUPLING TYPE CONTOURED DIAPRHRAGM FLEXIBLE
MANUFACTURER LUCAS (BENDIX) MODEL 74T522A SIZE
ASSEMBLY DRAWING NO.

NOTE: INFORMATION TO BE COMPLETED: ○ BY PURCHASER; ☐ BY COUPLING MANUFACTURER.

OPERATING CONDITIONS

● POWER TRANSMITTED (kW): NORMAL 36754 MAXIMUM CONTINUOUS
● SPEED (RPM): MINIMUM ALLOWABLE 3688 NORMAL 4610 MAXIMUM CONTINUOUS 4873 TRIP 5360
○ TORQUE (N-m): MAXIMUM CONTINUOUS MAXIMUM TRANSIENT
○ FREQUENCY OF TRANSIENTS (EVENTS/TIME) (2.1.1)
○ ADDITIONAL ANF EXCITING FREQUENCY (2.3.4.C)
● MINIMUM REQUIRED EXPERIENCE FACTOR, SF (2.1.1) 1.75 ☐ ACTUAL SF
● AMBIENT TEMPERATURE (°C) MAXIMUM 37 MINIMUM 21
ENVIRONMENT (2.6.5) (2.1.6) ○ DUST ○ HYDROGEN SULFIDE ● SALT LADEN ATMOSPHERE

LUBRICATION

○ CONTINUOUS (2.2.7 & 8) ○ FILTRATION MICRONS
○ BATCH (2.2.7) ● NONLUBRICATED
○ GREASE-PACKED
○ OILFILLED

○ VISCOSITY (cP) @ °C
○ PRESSURE (kPa) (BAR) NORMAL MAXIMUM
○ TEMP. (°C) NORMAL MAXIMUM
☐ FLOW (m³/h) NORMAL MAXIMUM

COUPLING DATA

● SHAFT SEPARATION AT OPERATING TEMP. (mm B.S.E.) (1) THERMAL GROWTH (mm) FROM °C AMBIENT
○ MOTOR ROTOR FLOAT (mm) ○ LIMITED END FLOAT (mm) (2.2.3) ○ ELECTRICALLY INSULATED (2.1.12)
○ MARINE TYPE REQUIRED (2.2.2) ○ FLEX-HUB TYPE REQUIRED (2.2.2)
○ REQUIRED MISALIGNMENT CAPABILITY (2.1.2)
STEADY STATE: ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL (mm)
TRANSIENT: ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL (mm)
○ DYNAMIC BALANCE (2.5.2.1) ● COMPONENT BALANCE & ASSEMBLY CHECK BALANCE (2.5.2.1) (2.5.2.3)
○ ASSEMBLY BALANCE (2.5.2.4) ○ REPEATABILITY TEST AFTER ASSEMBLY BALANCE (2.5.2.5)
○ RESIDUAL UNBALANCE CHECK OF COMPONENTS (2.5.2.2)
☐ DESIGN RATING (kW/100 RPM) (2.1.1) ☐ MAXIMUM CONTINUOUS TORQUE (N-m) (5.2.5.6c)
☐ BOLTING TORQUE (N-m) (2.1.11.5) ☐ LUBRICATED ☐ DRY
☐ MAXIMUM ALLOWABLE MISALIGNMENT (5.2.5.6a)
STEADY STATE: ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL (mm)
TRANSIENT: ANGULAR (DEG.) PARALLEL OFFSET (mm) AXIAL (mm)
☐ MAXIMUM ALLOWABLE RESIDUAL UNBALANCE (g-mm) DRIVER END DRIVEN UNIT END
☐ MAXIMUM ACTUAL RESIDUAL UNBALANCE (g-mm) DRIVER END DRIVEN UNIT END
☐ TORSIONAL STIFFNESS (kg-mm/RAD) (5.2.5.2b) WR²(kg-m²) (5.2.5.2b)
☐ FLEXIBLE-ELEMENT COUPLING INITIAL DEFLECTION (mm) ☐ PRESTRETCH ☐ COMPRESSION
☐ FLEXIBLE-ELEMENT COUPLING CALCULATED AXIAL NATURAL FREQUENCY
○ TEST OF ANF (4.4.1) ☐ ACTUAL ANF

(1) DUPLICATE OF TRAIN - G

NOTE: VENDOR TO COMPLETE THE DATA IN THE BLANK CELLS

IKPT

KELLOGG

COUPLING (API 671-2ND)
DATA SHEET
SI UNITS

JOB NO. 86098/6413 ITEM NO. HHK-2
REVISION NO. 3 DATE 24 MARCH 1997
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COUPLING DATA (CONTINUED)

● REQUIRED CURVES (5.2.4.1) (5.2.4.2)
- ● AXIAL FORCE VS AXIAL DEFLECTION
- ● AXIAL FORCE VS TORQUE
- ● BENDING MOMENT VS AXIAL DEFLECTION
- ● BENDING MOMENT VS ANGULAR MISALIGNMENT
- ○ WINDAGE LOSS VS COUPLING SPEED

MATERIALS (2.6)

	DRIVE END MATERIALS	DRIVEN END MATERIALS
HUB/FLANGE		
SPACER		
SLEEVE		
FLEXIBLE-ELEMENT		
FLEXIBLE-ELEMENT GUARD		
BOLTS		
NUTS		

● FLEXIBLE-ELEMENT PROTECTIVE COATING (2.6.4) ● VENDOR STANDARD ○
☐ MATERIAL AND WELD HARDNESS TESTS REQUIRED (4.3.1.3)
☐ INTERNAL TEETH HARDNESS (ROCKWELL C) (2.2.6) DRIVE END ACTUAL DRIVEN END ACTUAL
☐ EXTERNAL TEETH HARDNESS (ROCKWELL C) (2.2.6) DRIVE END ACTUAL DRIVEN END ACTUAL

COUPLING HUB MACHINING

	DRIVE END	DRIVEN END
● TYPE (INTEGRAL, CYLINDRICAL, TAPER) (2.1.4) (2.1.8.1)	INTEGRAL	
● KEYED OR HYDRAULICALLY FITTED (2.1.8.1)		HYDRAULIC FIT
○ TAPER (1 DEG. I.A., 1/2 IN. PER FT., 3/4 IN. PER FT.) (2.1.9.1.1) (2.1.9.1.2) (2.1.9.1.3)		
☐ KEYWAY DIMENSIONS AND NUMBER (2.1.9.2)		
☐ NOMINAL BORE DIAMETER		
☐ INTERFERENCE FIT (mm) MAX./MIN. (2.1.8.1)	/	/
○ PULLER HOLES		
○ BALANCING HOLES (2.5.3)		

COUPLING GUARD (APPENDIX E)

○ COORDINATOR (E.2.1)
● FLANGED CYLINDRICAL (E.4.1) ○ BASE MOUNT (E.3.2)
○ AIR TIGHT (E.4.7.1) ● OIL TIGHT (E.2.7)
● SPARK RESISTANT (E.2.9)
○ TRANSPARENT WINDOW FOR EACH OIL SPRAY POINT (E.4.8.4)
● VENT CONNECTION (E.4.3) (1)
- ○ 1 INCH NPT WITH FILTER-BREATHER
- ○ 1 INCH FLANGED, RATING & FACING

○ PURGE GAS ○ DRY AIR PURGE (E.4.4)
☐ PURGE CONNECTION SIZE & TYPE
☐ PURGE m³/h REQUIRED
○ OIL MIST COOLING ○ GAS COOLING
○ CONTRACT GUARD TO BE USED DURING SHOP TEST (E.4.5)
○ ADDITIONAL GUARD DETAILS (2.3.5)

ACCESSORIES

○ PRIME EQUIPMENT SUPPLIER TO FURNISH ONE SET OF PLUG AND RING GAGES (APPENDIX D)
○ DRILL TEMPLATE FOR INTEGRAL FLANGED HUBS BY (2.1.5)
- ○ COUPLING MANUFACTURER ○ PURCHASER

○ TWO-PIECE STOP RINGS BY COUPLING MANUFACTURER (5.3)

● HYDRAULIC INSTALLATION/REMOVAL TOOLING (TO INCLUDE HAND PUMP(S), PRESSURE GAUGE(S), FITTINGS AND HOSE(S)) BY (2.1)
- ○ PURCHASER ○ COUPLING MANUFACTURER

○ PULLER BY COUPLING MANUFACTURER (3.4)
● MOMENT SIMULATOR (2.1.7.1) FOR TURBINE DRIVER

APPLICABLE SPECIFICATIONS

● API-671, SPECIAL PURPOSE COUPLINGS
○

(1) DUPLICATE OF TRAIN - 6

○ COORDINATION MEETING ATTENDANCE REQD (6.2.2)

PREPARATION FOR SHIPMENT

● OUTDOOR STORAGE FOR MORE THAN 6 MONTHS (4.5.2)
● EXPECTED STORAGE TIME
SHIPPING: ○ DOMESTIC ● EXPORT (4.5.3)
STORAGE: ○ INDOOR ● OUTDOOR (4.5.2)
○ SEE COUPLING PURCHASE ORDER FOR PRESERVATION, BOXING, AND SHIPPING INSTRUCTIONS